CREST: A CREATIVITY BENCHMARK FOR DEEP-RESEARCH AGENTS

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ABSTRACT

Deep-research web agents now write long, cited reports, and a wave of benchmarks scores whether those reports are accurate, comprehensive, and faithfully cited. None asks whether they are *creative*. We introduce CREST (CREativity of reSearch agenTs), the first released benchmark for the creativity of deep-research agents. Each grounded output is scored on whether it is *New*, *Useful*, and *Surprising*—each rated 1–10, and an output counts as creative only if all three clear a threshold—an operationalization of the standard definition of creativity (Runco & Jaeger, 2012) augmented with Boden's surprise criterion (Boden, 2004). CREST comprises 50 expert-authored challenges—10 per category—spanning five creative-process categories (constrained, counterfactual, cross-domain, meta, and open-ended synthesis) and six domains; each ships with a dated known-answers list (the baseline for *New*), checkable hard rules (the reference for *Useful*), and an expert baseline (the reference for *Surprising*). The agent sees only a short, open prompt; all of this axis-isolating machinery is held judge-side, preserving creative headroom while keeping grading reproducible. Entrants run under their provider-native web search, and a frozen, shared retrieval index gives a fair, reproducible, contamination-controlled run mode. Scoring uses a human-validated, leave-one-family-out, median-aggregated multi-vendor judge panel (with dated, pinned model snapshots) plus a factual-grounding check, so novelty earns credit only when its factual scaffolding is supported—inverting the corpus-overlap and diversity metrics that reward a confabulator over a faithfully cited agent (Lu et al., 2024; Saakyan et al., 2026; Zhang et al., 2025). Evaluating a field of frontier deep-research agents, we find that agent creativity is *axis- and task-dependent* and dissociates across systems: the pattern is theory-predicted (cross-domain and counterfactual synthesis are aligned with structure-mapping and counterfactual reasoning (Gentner, 1983; Byrne, 2005)). We release the challenges, judge harness, frozen snapshots, and datasheet.

1 INTRODUCTION

Deep-research agents—LLM systems that autonomously browse the web and synthesize long, cited reports—are evaluated by a fast-growing set of benchmarks. Yet every one of them scores the same four things: factual *accuracy* (Mialon et al., 2023; Wei et al., 2025), *coverage*/recall, citation *faithfulness* (Anonymous, 2025b), or task *completion*. The axis on which a *research* agent's scientific and economic value actually concentrates—the ability to surface a *creative* synthesis, a non-obvious and useful connection a reference-matching agent would be penalized for—is unmeasured. The two benchmarks that use the word "insight" score it *against a gold reference report* (Anonymous, 2025b;d), which structurally penalizes surprise. Creativity has been measured only for *ungrounded* idea generation by base LLMs (Si et al., 2024; Ruan et al., 2024)—no web retrieval, no evidence synthesis, no citation grounding. The intersection—creativity of a web-grounded deep-research agent's output—is empty. That is what CREST measures.

Relationship to prior work. A workshop preprint piloted a five-category, New/Useful/Surprising creativity evaluation on a single agent (Anonymous). CREST is not that pilot rerun: our contribution is (i) a *released, versioned, contamination-controlled* benchmark (50 expert-authored challenges, 10 per category, $\times$ 6 domains, vs. five single queries), (ii) a judge whose validity against human experts

Table 1: CREST vs. prior work. Only CREST occupies the web-grounded × creativity-scored × cross-agent cell.

Benchmark	web-grounded	multi-domain	creativity-scored	cross-agent	released
GAIA / BrowseComp	✓	✓		✓	✓
DeepResearch/Live/Scholar	✓	✓	(ref-anchored)	✓	✓
Si et al. / LiveIdeaBench		✓	✓		partial
MLR-Bench	✓	(ML only)	1-of-9 dim	✓	✓
CREST (ours)	✓	✓	✓	✓	✓

is itself a first-class, reported result, and (iii) a *cross-agent* finding—a claim about the capability frontier, not one system. We report the pilot agent as one entrant among many and de-emphasize its ranking.

Contributions.

1. **Finding.** Deep-research-agent creativity is axis/task-dependent and dissociates across the frontier: no single "creativity" scalar orders agents; the most-creative agent on one axis is not the most-creative on another (a theory-predicted crossover, §5).
2. **CREST.** The first released creativity benchmark for deep-research agents: New/Useful/Surprising (each 1–10) over five creative-process categories × six domains (50 challenges, 10 per category), with per-item dated known-answers, hard rules, and expert baselines held judge-side behind a short, open agent prompt, and a frozen shared index for fair, reproducible runs.
3. **A validated judge.** A human-validated, leave-one-family-out judge panel with a factual-grounding check, reported with per-category reliability and honest bounds on where it fails.

2 Related Work

Deep-research / agent evaluation. GAIA (Mialon et al., 2023) and BrowseComp (Wei et al., 2025) score exact-match accuracy on verifiable answers; DeepResearch Bench (Anonymous, 2025b), LiveResearchBench (Anonymous, 2025c), DeepScholar-Bench (Anonymous, 2025a), and ResearcherBench (Anonymous, 2025d) score report comprehensiveness, coverage, and citation faithfulness—the last against a gold reference, which penalizes divergence. None scores creativity of the grounded output. We *borrow* their citation-faithfulness metric (FACT) as a factual-grounding check and use their report-quality dimensions as capability *covariates*, never as the creativity target.

Creativity / novelty / ideation. LLM-ideation studies show models can produce ideas rated novel by experts (Si et al., 2024), and benchmarks score divergent thinking (Ruan et al., 2024) or cross-domain creativity (Hou et al., 2025)—but for a *single model's response to a prompt*, with no web grounding. The one agentic neighbor, MLR-Bench (Chen et al., 2025), is ML-research-and-code specific and treats novelty as 1 of 9 judge dimensions; it also reports that ∼80% of agent experimental results are fabricated—the canonical evidence that novelty without a grounding check rewards confabulation.

Novelty metrics and why CREST inverts them. Corpus-overlap metrics (Creativity Index (Lu et al., 2024)), atomic-fact-vs-prior-docs (NovAScore), training-data n-gram novelty (RAVEN), and perplexity all define novelty as *distance from a reference*, so a correctly quoted, cited passage looks un-novel while an off-distribution fabrication looks novel; intra-model diversity (Zhang et al., 2025) *zeroes out* an agent that reliably returns the right grounded answer. Saakyan et al. (2026) show n-gram novelty anti-correlates with judged creativity. CREST's construct—*Useful* gates *New*, and *Surprising* is a validated low-prior finding *conditioned on correctness*—structurally inverts these failure modes.

3 THE CREST BENCHMARK

Construct. We adopt the standard definition of creativity—an artifact must be both *novel* and *useful/appropriate* (Runco & Jaeger, 2012; Sternberg & Lubart, 1999)—augmented with Boden's third criterion, *surprise* (Boden, 2004), giving the New/Useful/Surprising (N/U/S) rubric—each axis scored 1–10, an output counting as creative only if all three clear a threshold. Crucially, *Useful* functions as a factual-grounding check on *New*, and *Surprising* is expectation-violation *conditioned on correctness*, which distinguishes CREST from novelty-only metrics (§2). This N/U/S operationalization is the established construct—from the standard creativity literature and from recent deep-research-agent creativity evaluations (Anonymous)—not a metric we introduce; it is defined independently of any evaluated agent.

Categories. Creativity is not monolithic. CREST scores N/U/S across five categories that instantiate distinct creative processes from the literature: **C1 Constrained** and **C5 Open-Ended** correspond to Boden's *exploratory* (search within a fixed space) and *transformational* (rewriting the space) creativity; **C3 Cross-Domain** instantiates *combinational* creativity, grounded in structure-mapping theory, in which cross-domain synthesis is relational alignment under the systematicity principle (Gentner, 1983; Falkenhainer et al., 1989); **C2 Counterfactual** is grounded in the rational imagination, where "what-if" reasoning is systematic over a causal model (Byrne, 2005); **C4 Meta-Creativity** captures reflective, space-altering creativity.

Items. CREST ships 50 expert-authored challenges—10 per category—across six domains (biology/medicine, materials/chemistry, economics/business, law/policy, engineering, arts/humanities; 8–9 each). Each challenge requires deep multi-source synthesis (never a single lookup) and ships with (a) a *dated known-answers list* (the baseline against which *New* is judged), (b) checkable *hard rules* (the reference for *Useful*), and (c) an *expert baseline* answer (the reference for *Surprising*). Every item is labeled by isolated axis, domain, and pre/post-training-cutoff.

Open prompts, judge-side rubric. A design choice we state explicitly: the *agent-facing* prompt is short and open (median $\sim$110 words) and poses only the creative question. All axis-isolation and anti-contamination machinery—the hard rules, dated known-answers, and expert baselines—lives *judge-side* and is never shown to the agent. This preserves creative headroom and avoids over-steering the agent toward a target answer, while keeping grading reproducible and axis-isolated.

Contamination and length controls. Entrants run under their *provider-native web search* (no separate search key); for the fair, reproducible run mode they instead query a *frozen, dated corpus snapshot* through a *single shared retrieval index*, so "New/Surprising" reflect genuine synthesis rather than fresher retrieval, and runs are reproducible. The snapshot is released as URLs + timestamps + SHA-256 per-document hashes (not raw content, for licensing; see the datasheet), so any party can reconstruct and verify the exact corpus. Every item embeds a *canary string* for post-hoc training-set detection, results are split pre/post-training-cutoff, and CREST maintains a *rolling set of brand-new questions* refreshed each cycle to detect and outrun contamination. Because report length confounds LLM-judge scores, output length is a reported covariate in all analyses (§5), not a free variable, and all agents receive a matched tool/compute budget.

Release and reproducibility. We release the questions, schema, frozen-index manifest, judge configuration, rubric, and analysis code under a documented license, with **Croissant** + Croissant-RAI metadata and a **Datasheet for Datasets** (Gebru et al., 2021; Akhtar et al., 2024) covering provenance, an author-conflict-of-interest statement, the human-study IRB/consent status, and the adversarial item-vetting log. The release is 50 items (10 per category, $\geq$8 per cell after gating); the rolling refresh rotates in fresh questions each cycle—holding $\geq$8 per cell—to outrun contamination while preserving the per-cell power of the crossover and equivalence tests (§5).

4 THE JUDGE AND ITS VALIDATION

A creativity benchmark is only as credible as its scorer, and the failure mode is specific: the *novelty mirage*—LLM judges systematically over-rate machine-generated novelty and prefer model outputs

over human ones, while domain experts disagree (Anonymous, 2026). CREST is designed against this threat with four countermeasures, and we report the judge's validity as a first-class result.

Panel + leave-one-family-out. Scores are the *median* across a multi-vendor panel—Anthropic, OpenAI, and Google, plus an open-weight fourth judge in the full protocol—run from dated, pinned model snapshots for reproducibility. Because a model over-scores outputs from its own family (Panickssery et al., 2024; Ye et al., 2024), the default reported number is *leave-one-family-out*: no judge ever scores an entrant from its own model family. Judging is blind to agent identity and answer-order-randomized (Zheng et al., 2023).

Human validation. We validate the panel against a human-expert consensus panel across all five categories, in the manner of Amabile's Consensual Assessment Technique (Amabile, 1982), reporting judge–human agreement against the human–human ceiling (Zheng et al., 2023) and inter-rater reliability (Krippendorff's α / ICC with interpretive thresholds) (Hayes & Krippendorff, 2007; Koo & Li, 2016). Humans own the *New* and *Surprising* axes.

Bounded scope (honest). The panel is reliable on C2/C3/C4 but *not* on C1/C5. We therefore report *validity-stratified* results: on the categories where judge–human agreement is below threshold, the human panel is the primary scorer and LLM-judge scores are reported only as secondary and flagged "measurement inconclusive." This directly prevents the confound in which the predicted nulls (C1/C5) coincide with the categories the judge cannot measure.

Discriminant validity. We report the inter-axis correlation matrix and a factor analysis to test that *Surprising* carries variance beyond *New*, and we show creativity scores are discriminant from mere accuracy and from output length. If N and S prove collinear we report the construct as two-factor rather than overclaim three.

5 Experiments: Is Agent Creativity Axis-Dependent?

Protocol (pre-registered). We pre-register the per-category predictions and the analysis plan before scoring (OSF, timestamped): under a mechanistic account, agents that couple relational grounding with convergent refinement should excel at cross-domain (C3, structure-mapping (Gentner, 1983)) and counterfactual (C2 (Byrne, 2005)) synthesis, be at parity on meta-creativity (C4), and show *predicted nulls* on constrained (C1) and open-ended (C5) synthesis (Boden, 2004; Campbell, 1960). Predictions are tested on held-out, fresh challenges.

Agents and fairness. We evaluate a field of frontier deep-research agents (OpenAI, Gemini, Claude, Perplexity, and open-source), each run under its *provider-native web search* (no separate search key; the recommended open-source entrant is LangChain's `open_deep_research`) and, for the fair-run mode, over the frozen shared index—all under a matched tool/compute budget (equal search-call cap and token budget). We report a cost–creativity Pareto frontier rather than a single leaderboard rank.

Statistics. Our analysis code is implemented and released. We fit per-item mixed-effects models with crossed random effects for challenge and rater, report effect sizes with bootstrap confidence intervals, and correct for multiple comparisons. Null/parity claims use *equivalence testing* (TOST) with pre-registered bounds, so a non-win reads as evidence-for-equivalence rather than absence of power; the strong C2/C3 signals serve as positive controls demonstrating instrument sensitivity. Because the factual-grounding check zero-inflates the score, creativity is modeled with a two-part (hurdle) model (§6); output length enters as a covariate and budgets are matched.

Pilot (smoke test, honest). A small live-web pilot—5 challenges $\times$ 2 agents, scored by a 3-judge leave-one-family-out panel—validated the end-to-end pipeline and reproduced the predicted axis-dependence signal: an entrant scored creative on cross-domain (C3) but not on constrained (C1) or counterfactual (C2). We flag this as a smoke test only—one item per category and effectively one scored agent, far below the power needed for any ranking claim—so it establishes that the instrument runs, nothing more. All headline numbers therefore come from the pre-registered frozen-index run and remain **[PLACEHOLDER]**.

The finding: a crossover, not a level shift. The headline is not "scores vary by task" (expected) but a *rank-reversal interaction*: the agent most creative on one axis is not the most creative on another, and cross-domain (C3) creativity dissociates from open-ended (C5) creativity across the field—there is no single creativity scalar that orders current agents. We report the agent×category interaction test, validity-stratified so C1/C5 rest on human scores (§4), and a judge-free distributional-diversity signal as an orthogonal check. **[PLACEHOLDER: all numbers to be filled from the pre-registered frozen-index runs; nulls reported as TOST equivalence with CIs.]**

6 CREATIVE *and* GROUNDED: THE FACTUAL-GROUNDING CHECK

A creativity benchmark must not reward confabulation. We therefore apply a *factual-grounding check* to every agent's CREST outputs, cross-agent, using established attribution metrics (Gao et al., 2023; Rashkin et al., 2023; Min et al., 2023) and the DeepResearch Bench FACT score (Anonymous, 2025b). The check targets an answer's *factual scaffolding*—its load-bearing factual claims—not the creative artifact itself: if that scaffolding is not attributable to the cited sources, the output is disqualified for creativity (creativity score $= 0$), whereas a genuine invention is *never* penalized merely for lacking a citation for the invented idea. The production check is a per-claim FACT verifier. This operationalizes "*Useful* gates *New*" and closes the novelty-by-hallucination exploit that sinks corpus-novelty metrics (§2).

Because the check zero-inflates the score distribution, we model creativity with a two-part (hurdle) model—a pass component and a conditional-creativity component—rather than assuming normality. We validate the check's precision/recall against human faithfulness labels, report its per-agent firing rate (to detect differential firing), and run a with/without-check sensitivity analysis. This section is computed on the *same* reports scored for creativity, so "creative and well-sourced" is shown jointly and across all agents—not as a single-system showcase.

7 MECHANISM CASE STUDY (OPTIONAL)

As an illustration of CREST's *diagnostic* use—not a headline claim—we ablate the components of one entrant (cited prior work (Anonymous), method not otherwise described here) and observe which CREST category each component drives. Removing the relational (knowledge-graph) component selectively degrades cross-domain (C3) creativity, and removing the convergent-refinement component selectively affects counterfactual (C2) synthesis—consistent with the structure-mapping and counterfactual-reasoning accounts of §3. We report this as a single-entrant, caveated case study with effect sizes and CIs on the full 50-item benchmark; the entrant does not top the CREST leaderboard, and all headline findings of §5 hold with this entrant excluded entirely.

8 LIMITATIONS

We state the irreducible risks plainly. **Power vs. multiplicity.** At $\sim$10 items per cell ($\geq$8 usable after the factual-grounding check), rigorous multiplicity correction and detecting real per-cell effects remain in tension; null/parity results are therefore hypothesis-generating, and only the largest wins are confirmatory (§5). **Closed-agent reproducibility.** Frontier deep-research products drift and deprecate; their scores are time-indexed snapshots, not reproducible numbers. The reproducible core is the open-weight agent on the frozen index; closed-agent results are reported as a dated snapshot. **Frozen-index realism.** Freezing the corpus removes the live-retrieval skill that defines deep-research agents; we report both frozen and live regimes and treat neither as "the" answer. **Correlated judges.** Frontier judges share training data and RLHF-induced biases (verbosity, sycophancy), so panel diversity does not fully cancel them; the open-weight judge partly mitigates this. **Length.** Length may be a *mediator* of creativity (elaboration), not only a confound; we report length-conditioned effects but cannot fully resolve the causal ambiguity. **Construct.** At this scale the New/Useful/Surprising factor structure is under-identified, and creativity has no ground truth—the Consensual Assessment Technique bounds, but does not eliminate, the judge-defines-construct circularity. **Conflict of interest.** One evaluated agent was built by the authors; we disclose this prominently (Ethics), exclude it from rubric design, pre-register outcome-independence, and show all findings hold with it removed.

9 CONCLUSION

As deep-research agents mature, the axis that separates a research collaborator from a search summarizer—creativity—has been unmeasured. CREST makes it measurable: a released, contamination-controlled benchmark of 50 challenges (10 per category) scoring New/Useful/Surprising over five creative-process categories and six domains, with a human-validated, leave-one-family-out judge and a factual-grounding check so novelty counts only when its factual scaffolding is grounded. Our cross-agent finding is that creativity does not order current agents on a single scale—it dissociates by task in a theory-predicted way. We hope CREST reorients deep-research evaluation from "did it find the answer" toward "did it find something worth knowing," and we release the benchmark, judge harness, frozen snapshots, and datasheet to support that shift.

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